

DeepLearning.AI

Improving agentic applications

Module 5

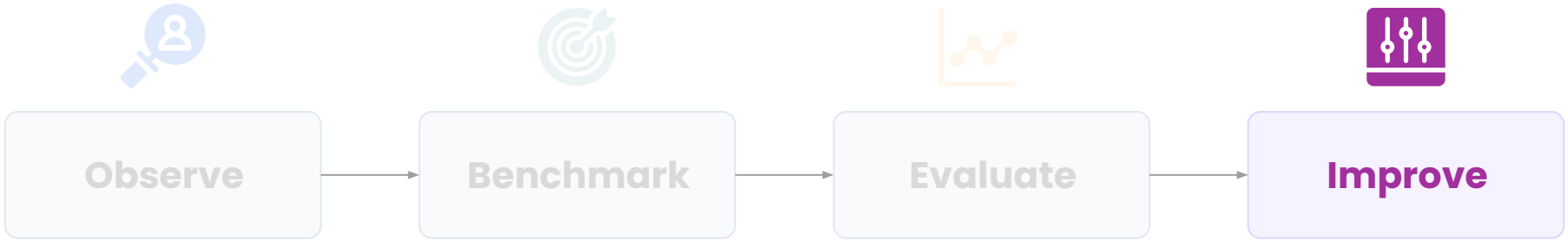


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Planning an improvement test

Module 5 · Video 1

Plan an improvement test



Target behavior → Candidate change → Stable comparison

Experience is not current handling ability

Failed matches



Experience with large dogs



Buddy, 130 lb pulls strongly on walks



Agent: approved

Current handling ability: not assessed



Evaluator: rejected

Physical capacity wasn't assessed

Experience with large dogs was treated as sufficient

The dog's pulling strength wasn't considered

Find a behavior worth improving



QA

Past experience treated as enough evidence

Assess current physical handling ability

Physical capacity wasn't assessed

Experience with large dogs was treated as sufficient

The dog's pulling strength wasn't considered

Write an eval for a capability

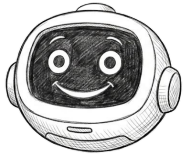
New capability



Pamela

Don’t schedule me for walks after six

Expected to fail at first → Target to develop



Agent

Regression eval:
preserve an ability

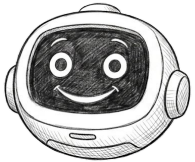
Capability eval:
develop an ability

A plausible answer can add unsupported claims

How should I care for the merino wool sweater?



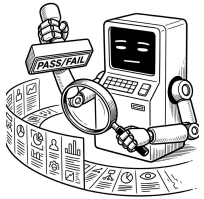
Machine washable on wool cycle



Agent

Here is what to keep in mind:

- Machine washable on wool cycle
- Cool-water machine wash
- Wool-specific detergent
- Cedar blocks for moths



Faithfulness error

Unsupported product claims

Test a grounding hypothesis



Agent
Baseline

Use tools when helpful.
Give concise answers.



Agent
Grounding rule

Use tools when helpful.
Give concise answers.
Never state a product detail unless a tool result supports it.

Machine washable on wool cycle.
Use only retrieved care information

A hypothesis, not a proof

Change one thing, keep the comparison stable



Agent
Baseline



Agent
Grounding rule

Hypothetical: score +8 points

Different cases?

Different evaluators?

Different model?

Same cases · Same evaluators · Same model
Same other configuration

Test capability and protect known-good behavior

Capability cases	Known-good cases
Merino wool care	Sizing questions
Other unsupported product claims	Product-information requests
Does it improve?	Does it still work?
Capability	Regression protection

Set the bar before you iterate

Good enough for this application?

Value of improvement

Safety / policy reliability

Remaining regressions + effort

Target behavior → Candidate change → Stable comparison

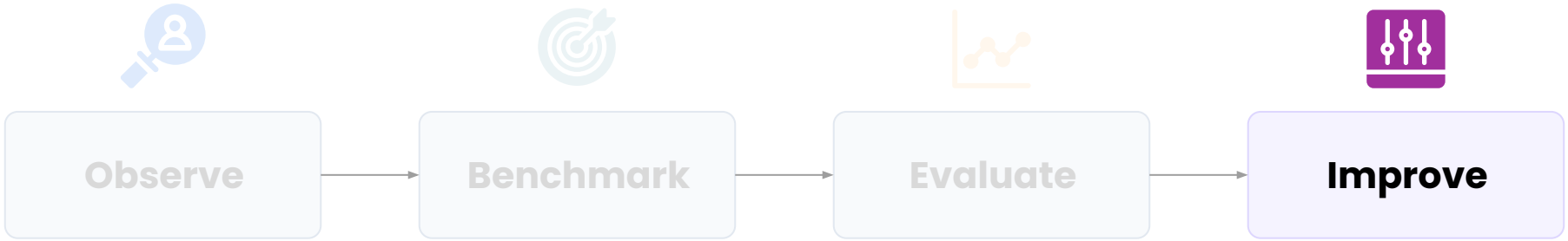


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Comparing versions and assessing tradeoffs

Module 5 · Video 2

Run the improvement comparison



Save cases → Run both versions → Inspect changes

Capability

Regression

Save the cases you will reuse



Faithfulness traces: unsupported product details

Northwind faithfulness failures	Known-good cases
Merino wool care	Sizing questions
Other unsupported product claims	Product-information requests
Target improvement	Protect working behavior

Run the same experiment on both versions



Agent
Baseline



Agent
Grounding rule

Same model · Other instructions · Evaluators

Dataset

Merino care

Known-good

Apply the same task to every example



Builder

Task

One question → Northwind → Response

Experiment

Results + Traces

Evaluators



Faithfulness



Code evaluator

Keep the care answer grounded

How should I care for the merino wool sweater?



Machine washable on wool cycle



Agent

Baseline	Grounding rule
Here is what to keep in mind: Machine washable on wool cycle Cool-water machine wash Wool-specific detergent Cedar blocks for moths	The sweater can be machine washed on the wool cycle.
Faithfulness: unsupported claims	Faithfulness: improved in this case

The answer stayed correct; the trajectory changed

Action / tool call | **Trajectory** | Session | System outcome



Agent
Baseline



Agent
Grounding rule



What size should I get?

Known-good
Sizing information already available

Same correct answer

Trace · same question + context

Uses available information
No additional lookup

Trace · same question + context

Redundant product lookup
One additional lookup



Hypothesis: fresh evidence required?

Read failure categories separately

Unsafe matches ↓



First-time volunteer



Buddy, 130 lb Experienced handlers only

Eligible volunteers rejected ↑



First-time volunteer



Daisy, 45 lb, calm

One overall score can hide both movements

Decide whether the tradeoff is worth it



Agent
Baseline

Faithfulness: unsupported

Final-answer correctness: correct

Redundant tool use: none

Risk of unsupported claims



Agent
Grounding rule

Faithfulness: grounded

Final-answer correctness: correct

Redundant tool use: extra lookup

Tool cost + latency



Faithfulness



Code evaluator

Next hypothesis: use available support; look up only when missing

Weigh value, severity and effort



QA

Large cluster → Smaller group → Rare cases

Rare safety failure can still matter

Another iteration?

- Value of improvement
- Severity of remaining failures
- Engineering effort

Let the comparison guide the next decision

Required performance
Acceptable regressions?

Good enough for now

Another round



Open changed traces



Evidence for the
next intervention

Data

Prompt

Tools

Workflow

Model

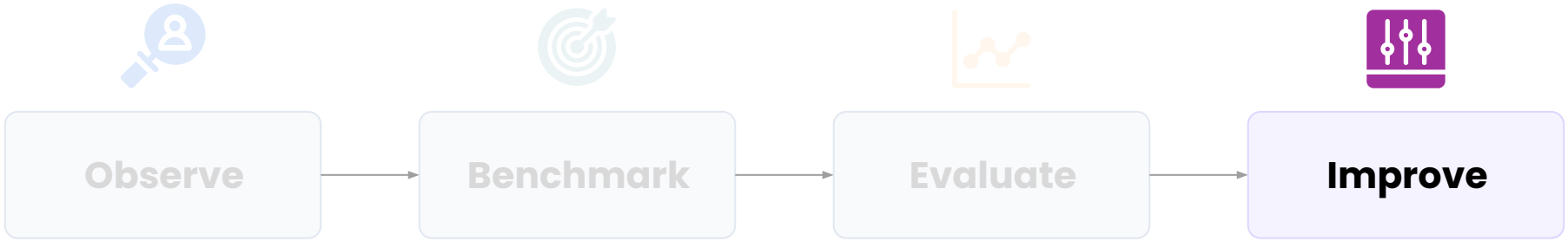


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Choosing where to intervene

Module 5 · Video 3

Choose the intervention from the evidence



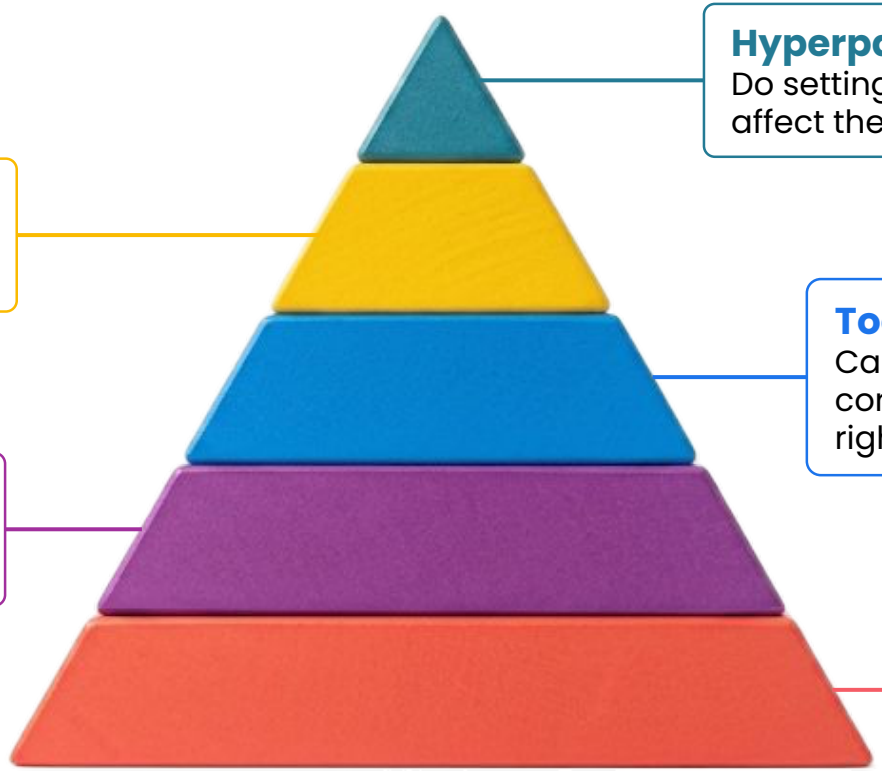
Northwind: extra sizing lookup

Locate the failure → Consider interventions → Start from evidence

Prompt · Data · Tools · Workflow · Model

Where does the failure show up?

Start at the simplest plausible layer



Hyperparameters
Do settings materially affect the behavior?

Model
Is the model capable enough for the task?

Tools & workflow
Can the agent act correctly and in the right order?

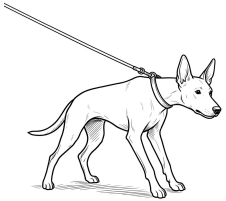
Prompting
Are the instructions clear enough?

Data quality
Does the agent have the right information?

Start with the data



Maria
First-time volunteer



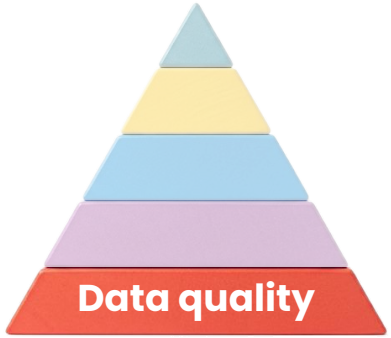
Atlas
pulls hard



Agent
Matches Atlas with Maria

Experienced handler required: blank

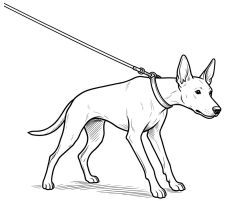
Prompt workaround?
“Make sure you read the behavior notes before deciding.”



Populate the requirement at the source



Maria
First-time volunteer

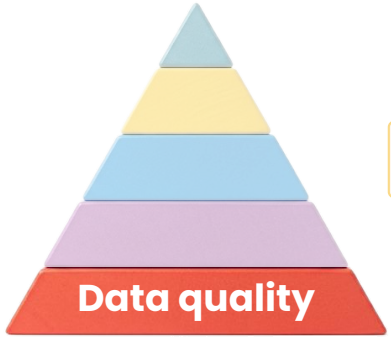


Atlas
pulls hard



Data-entry / processing rule

Experienced handler
required: yes



Same eval again: did the failure disappear?

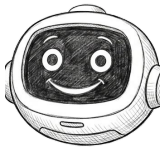
Clarify the requirement in the prompt

Can you book me with Pepper for Saturday afternoon?



Pamela
Can safely handle large dogs
Record complete

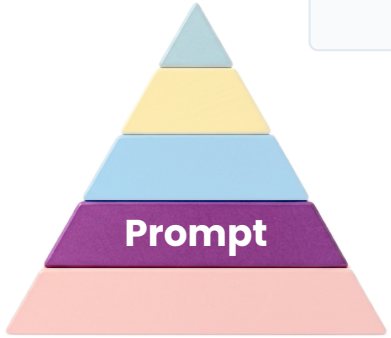
Great, you're booked with Pepper for Saturday at 2 PM.



Agent



Pepper, 100 lb



Retrieve record Pamela → Eligibility check missing

↓

Match volunteer and dog: Pepper → Create booking Saturday, 2 PM

Does the prompt require the check?

Clarify the requirement in the prompt

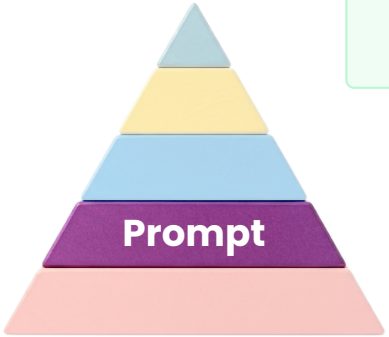
Always check eligibility before scheduling an outing



Pamela
eligible → continue

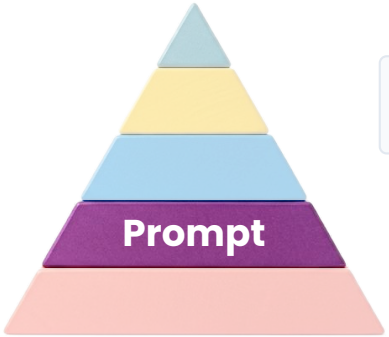


Volunteer 2983
not eligible → stop



Test beyond the examples you gave

Prompt examples		Different test cases	
Large dog	Large dog	Small dog	Medical restriction
		Different wording	



Learn the requirement, not a narrow pattern

Give tools a clear contract



Agent

Prompt requires an eligibility check

No match.

What happens next?

Tool contract: when to use it + possible results + next action

eligible
continue

not eligible
stop

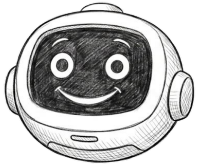
no match
ask for identifying
information



Make the required check part of the workflow



Pamela

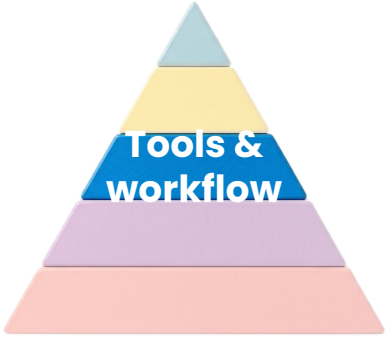


Agent



Pepper, 100 lb

Receive request → Check eligibility → Schedule



Still skipped despite clear instructions?

Every added step has a cost



What size should I get?

Known-good
Sizing information already available

Northwind sizing

One additional lookup

Latency

API cost

Another failure point

Required safety step

Optional step

Test removing an optional step → Same evals

Consider the model after local layers

Local interventions | Data ✓ | Prompt ✓ | Tools & workflow ✓



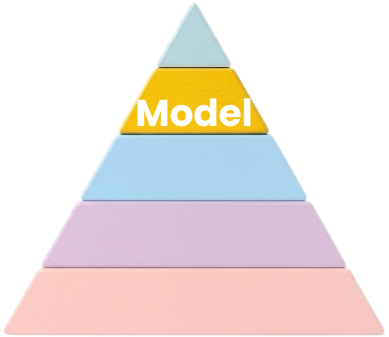
Agent
Current model



Agent
Candidate model

Many requests affected
Latency · Cost

Same evaluation suite

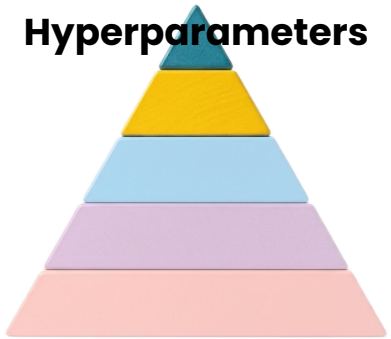


Capability cases	**Known-good cases**
Merino wool care	Sizing questions
Other unsupported product claims	Product-information requests

Tune sampling when variability is the problem

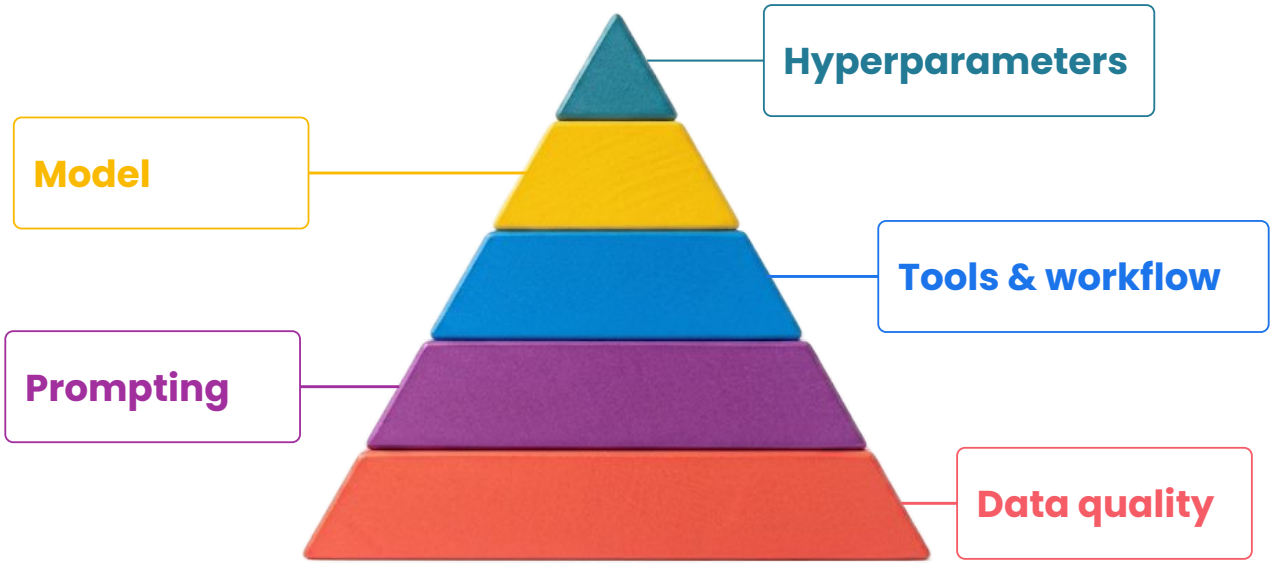
If the model exposes these settings

Temperature	Sampling	Output variability
Missing field	Undefined no match	Missing eligibility instruction



Match the setting to a variability problem

Match the change to the evidence



Where does the evidence point?

Missing or inconsistent information	Unclear requirement	Execution difficulty	Task still too hard	Sampling problem

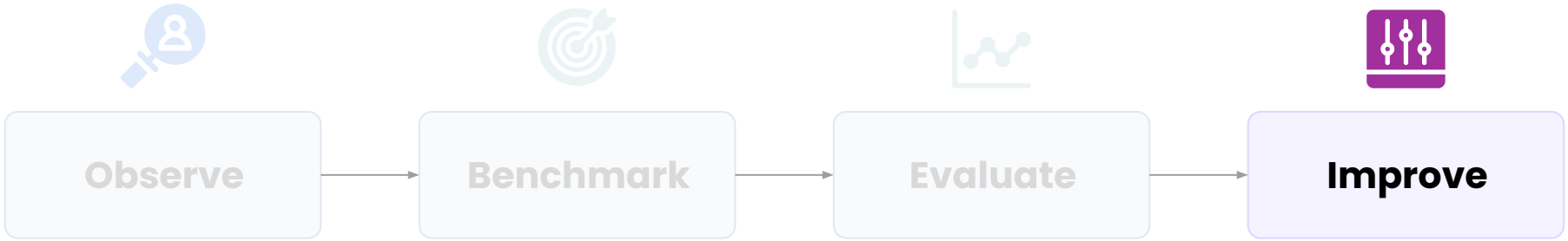


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Refining a change after a regression

Module 5 · Video 4

Refine a change after a regression



Northwind: first grounding rule

Care: grounded → Sizing: extra lookup → Evidence for the next change

The answer stayed correct; the trajectory changed

Decision to make an extra lookup



Agent
Baseline



Agent
Grounding rule



What size should I get?

Known-good
Sizing information already available

Trace: same question + context

- Uses available information
- No additional lookup

- Redundant product lookup
- One additional lookup

Same correct answer

Same request

Same available information

Product tool works

Investigate the prompt first

Sizing information already available

Data: present

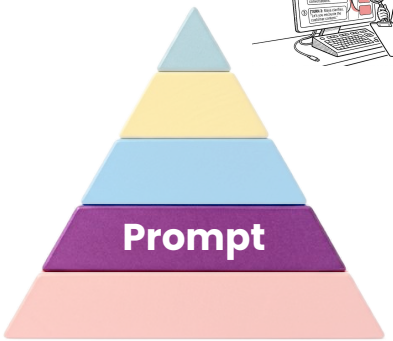
Tool: working, unnecessary lookup

Changed instruction



Does grounding imply a fresh lookup?

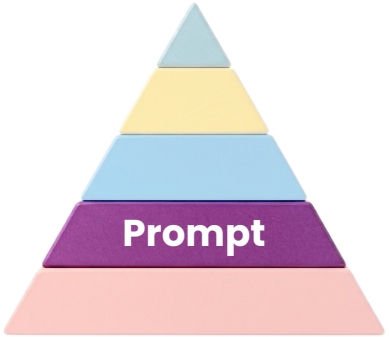




Make the ambiguity visible



Care: supported information only



Never state a product detail unless a tool result supports it.

Earlier tool result
Current context

Retrieve evidence again?

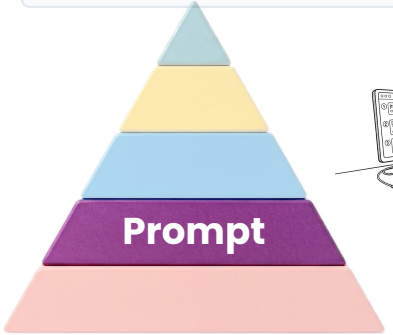
Clarify when another lookup is needed



Candidate refined rule

Never state a product detail unless supported by the information available. If the information available is insufficient, call the product tool to get additional information

Use available catalog information → Lookup if insufficient → Preserve grounding





Hypothesis → Run another experiment

Compare three versions on the same benchmark



Baseline



Grounding rule



Refined rule

Same benchmark

Capability cases

How should I care for the merino wool sweater? Machine washable on wool cycle.

Known-good cases

What size should I get? Sizing information already available



Faithfulness



Code evaluator

Faithfulness: still grounded?

Redundant tool use: removed?

Other known-good behavior · Tool use · Latency + cost

If the refined rule works as hoped

Hypothetical outcome

Agent · Baseline	Agent · Grounding rule	Agent · Refined rule

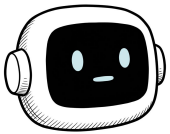
Same benchmark

Capability cases	Known-good cases		
How should I care for the merino wool sweater? Machine washable on wool cycle.	What size should I get? Sizing information already available	 Faithfulness	 Code evaluator
Care: Unsupported claims	Care: Grounded	Care: Grounded	
Sizing: No additional lookup	Sizing: One additional lookup	Sizing: No additional lookup	

If other cases stay stable, keep the improvement

Different results guide the next iteration

Alternative outcomes



Agent
Refined rule

Alternative A

Fewer lookups,
unsupported claims return

Alternative B

Sizing fixed,
new problem elsewhere



Inspect changed cases → More specific candidate → Test again

Keep the cases that protect what you learned

Capability cases	**Regression protection, once reliable**
Other unsupported product claims Not yet reliable	How should I care for the merino wool sweater?
Former role: Capability target	**Tool:** Machine washable on wool cycle. **Expected:** The sweater can be machine washed on the wool cycle.
Unfixed failures: learn	Fixed behavior: preserve



Failure → Evidence-based change → Compare and inspect

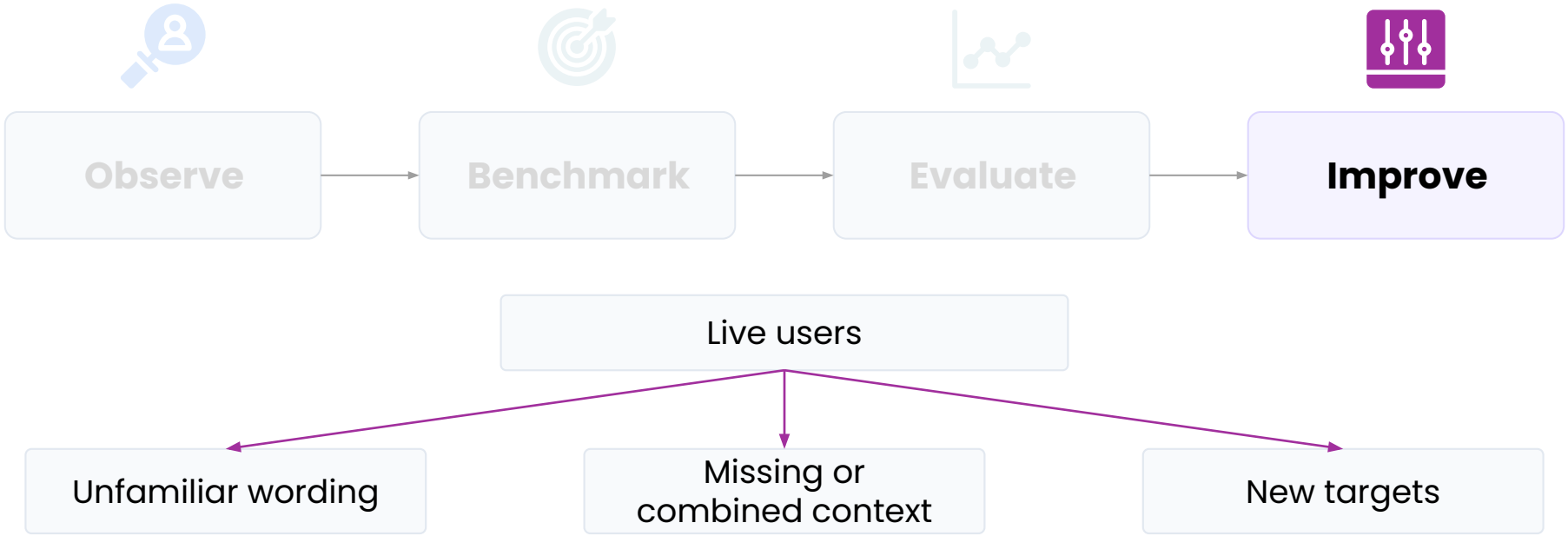


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Improving agents after deployment

Module 5 · Video 5

Find the next targets in production



Production changes how requests are phrased

Offline wording	Production wording	
I have limited mobility.	I can only walk for about ten minutes.	Is Pepper free Saturday, and if not, can I switch to Sunday?

Right answer
2 → 14 seconds

Changed shelter policy

New dog record fields

The benchmark covers only the situations it contains

Keep offline tests and add production evidence

Repeatable offline benchmark

Capability cases	**Known-good cases**
Mobility wording	Working scheduling requests

Candidate change → Same benchmark → Before release

Automated checks
Product-specific release rule



Code evaluator



QA

Production monitoring

Manual review

User feedback

A/B testing

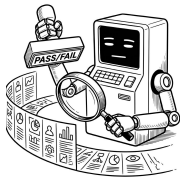
Monitoring locates; review explains

Production monitoring

Policy flags

Tool use

Response latency



Policy evaluator

Where should I look?

Manual review



Inspect live traces

Hypothetical: 37 eligibility failures

Skipped check?

Wrong argument?

Ignored result?

Wrong conclusion?

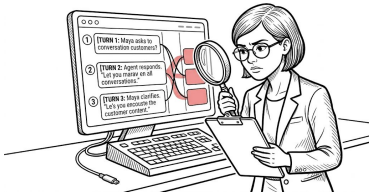
Count → Trace → Pattern

Use feedback to choose what to inspect

User feedback

Explicit: thumbs-down

Behavioral: abandon the conversation



Inspect the interaction

A signal points to an interaction; it does not diagnose the cause

Compare versions on real traffic

Real traffic





Agent **Current**	Agent **Revised**

Policy failures	Follow-up turns	Accuracy	Latency
Monitoring: where	Review: what happened	Feedback: user problems	A/B: useful in traffic

Different questions, complementary evidence

Small inefficiencies grow with traffic

What size should I get?

First grounding rule: Never state a product detail unless a tool result supports it.



Agent · Baseline

Uses available information
No additional lookup



Agent · Grounding rule

Redundant product lookup
One additional lookup

Care: grounded

Sizing: extra lookup

Hypothetical: One extra call → Thousands of times a day

Latency | API cost | Tokens | Handoffs | Repeated turns

Correctness + Operational efficiency

Locate the missed constraint in a live trace



Short-walk volunteer

I can only walk for about ten minutes. Is Pepper free on Saturday?



Pepper, 100 lb Energetic

Production trace

Availability checked

Outing scheduled

Walking constraint not considered



Agent

Recognize constraint before matching



QA

Save the original wording and expected behavior



Short-walk volunteer

I can only walk for about ten minutes. Is Pepper free on Saturday?



Pepper, 100 lb
Energetic

Recognize constraint → Check compatibility → Schedule only if appropriate



QA

Saved case:
Original message +
Expected behavior



Builder

Targeted change →
Run the
whole benchmark

Older mobility wording

Working scheduling
requests

Feed production evidence back into offline evals



Short-walk volunteer

I can only walk for about ten minutes. Is Pepper free on Saturday?

If the constraint is handled and the rest stays healthy

I'm fine with short walks, but nothing strenuous.



QA



LLM judge



Builder

1 Production trace

2 Expected behavior

3 Saved case

4 Offline comparison

5 Release & monitor

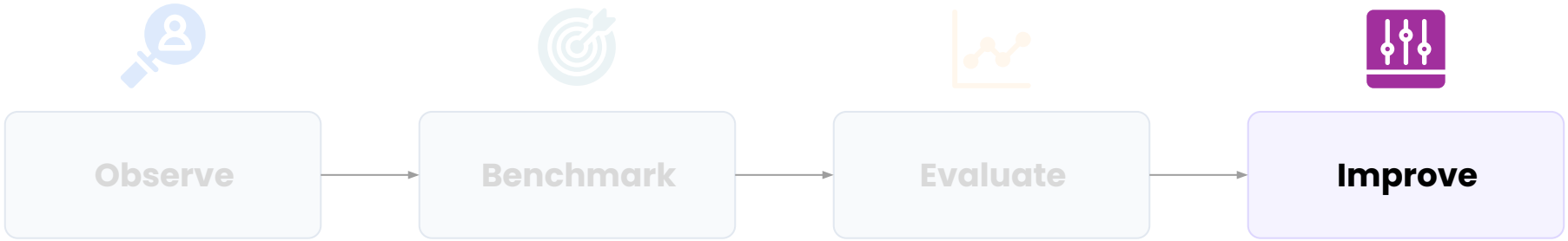


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Turning production failures into regression tests

Module 5 · Video 6

Bring production failures into the improvement loop



Hypothetical offline policy compliance: 96%

Which cases?

Which wording?

What is missing?

A high score covers the cases you tested

Hypothetical offline score: 96%

Offline examples

Refund order 1842

I want a refund for order 2271

No similar wording in this offline set

Production wording



Customer C103

I bought this last week and it doesn't fit. Can I get my money back?

No "refund" keyword

96% describes tested cases, not unseen wording

The failure starts before the refund tools

Action / tool call	Trajectory	Session	System outcome



Customer C103

I bought this last week and it doesn't fit. Can I get my money back?



Agent

I'm sorry it didn't work out! Would you like to fill out our feedback form?

Trace · Missing actions

```
get_customer: Not called
```

```
order_lookup: Not called
```

```
process_refund: Not called
```



QA

Refund intent not recognized

→

Original message
→ Offline dataset

Save exactly as it appeared

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arize

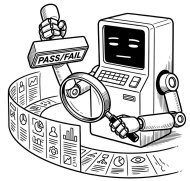
Save expected behavior with separate checks



Code evaluator

Expected: recognize refund intent

Follow policy
Required checks before processing



Policy evaluator

Tool evaluators
Required lookups and actions present, in order?

Policy evaluator
Refund rules followed?



QA

Repeatable test case
Original message + Expected behavior + Evaluators

Investigate the entry to the refund workflow

Refund intent not recognized

Routing instruction

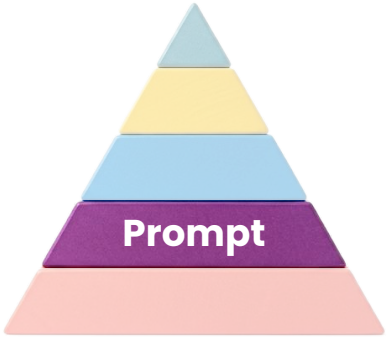


Builder

Before any refund tool call

Examples of intent

Clarify when to enter
Vary refund intent wording





QA

A concrete hypothesis, not the final fix

Run the expanded benchmark

Expanded benchmark

New capability case

I bought this last week and it doesn't fit. Can I get my money back?

Known-good protection

Refund order 1842

I want a refund for order 2271.

How should I care for the merino wool sweater?



Code evaluator



Policy evaluator



Agent · Updated

Check the new case and what was already working

I bought this last week and it doesn't fit. Can I get my money back?

→ Expected: recognize refund intent and follow the refund workflow

Questions for this comparison

Older refund requests: still working?

Other known-good cases: Unintended change?

Future versions must preserve this behavior

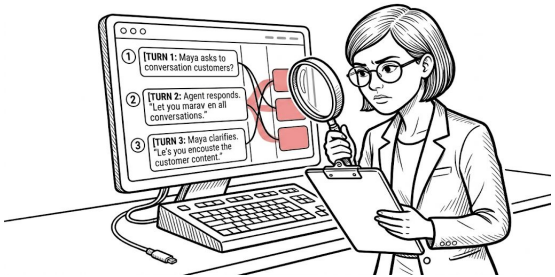
Case origins:

- Development failure
- New capability
- Introduced regression
- Production

Take what you learned and add it to the next eval

Feed production evidence back into offline evals

1. Get eval results
2. Choose what to improve
3. Make a targeted change
4. Compare versions
5. Inspect and refine



The evaluation cycle

